

Commerce XI Maths App

Theory (Pages 10-15) - Detailed Answers with Explanations

Total questions: 35 | +2 correct | -1 wrong

This answer guide follows the question order used in the app.

Question 1 - Theory Q1

In an ordered pair (a, b) , why is the order important?

Options

- A. (a, b) and (b, a) are generally different
- B. The components must be equal
- C. Only the second component is used
- D. The components must be natural numbers

Correct answer: (a, b) and (b, a) are generally different

Explanation and steps

1. The governing definition or property gives: (a, b) and (b, a) are generally different.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is (a, b) and (b, a) are generally different.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 2 - Theory Q2

In the ordered pair (a, b) , a is called the:

Options

- A. First component
- B. Second component
- C. Image only
- D. Co-domain

Correct answer: First component

Explanation and steps

1. The governing definition or property gives: First component.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is First component.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 3 - Theory Q3

Two ordered pairs (a, b) and (c, d) are equal if and only if:

Options

- A. $a = c$ and $b = d$

- B. $a = d$ and $b = c$
- C. $a = b$ and $c = d$
- D. $a + b = c + d$

Correct answer: $a = c$ and $b = d$

Explanation and steps

1. The governing definition or property gives: $a = c$ and $b = d$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $a = c$ and $b = d$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 4 - Theory Q4

$(a, b) = (b, a)$ if and only if:

Options

- A. $a = b$
- B. $a \neq b$
- C. $a = 0$
- D. $b = 0$

Correct answer: $a = b$

Explanation and steps

1. The governing definition or property gives: $a = b$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $a = b$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 5 - Theory Q5

The Cartesian product $A \times B$ is the set of:

Options

- A. All ordered pairs (a, b) where $a \in A$ and $b \in B$
- B. All unordered pairs from A and B
- C. Elements common to A and B
- D. All subsets of $A \cup B$

Correct answer: All ordered pairs (a, b) where $a \in A$ and $b \in B$

Explanation and steps

1. The governing definition or property gives: All ordered pairs (a, b) where $a \in A$ and $b \in B$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is All ordered pairs (a, b) where $a \in A$ and $b \in B$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 6 - Theory Q6

How is the Cartesian product of A and B read?

Options

- A. A cross B
- B. A union B
- C. A intersection B
- D. A related to B

Correct answer: A cross B

Explanation and steps

1. The governing definition or property gives: A cross B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is A cross B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 7 - Theory Q7

If $A = \{1, 2\}$ and $B = \{a, b\}$, which ordered pair belongs to $A \times B$?

Options

- A. $(2, b)$
- B. $(a, 1)$
- C. $(b, 2)$
- D. (a, b)

Correct answer: $(2, b)$

Explanation and steps

1. The governing definition or property gives: $(2, b)$.
2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is (2, b).

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 8 - Theory Q8

If $A = \emptyset$ or $B = \emptyset$, then $A \times B$ is:

Options

- A. $\emptyset$
- B. A
- C. B
- D. $\{\emptyset\}$

Correct answer: $\emptyset$

Explanation and steps

1. The governing definition or property gives: $\emptyset$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $\emptyset$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 9 - Theory Q9

For finite sets A and B, which formula is correct?

Options

- A. $n(A \times B) = n(A) \times n(B)$
- B. $n(A \times B) = n(A) + n(B)$
- C. $n(A \times B) = n(A) - n(B)$
- D. $n(A \times B) = 2^{nA}$

Correct answer: $n(A \times B) = n(A) \times n(B)$

Explanation and steps

1. The governing definition or property gives: $n(A \times B) = n(A) \times n(B)$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $n(A \times B) = n(A) \times n(B)$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 10 - Theory Q10

A relation R from a non-empty set A to a non-empty set B is:

Options

- A. Any subset of $A \times B$
- B. Always equal to $A \times B$
- C. Any subset of $A \cup B$
- D. A subset of $B \times A$ only

Correct answer: Any subset of $A \times B$

Explanation and steps

1. The governing definition or property gives: Any subset of $A \times B$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Any subset of $A \times B$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 11 - Theory Q11

If $(x, y) \in R$, the notation xRy is read as:

Options

- A. x is related to y under R
- B. x is equal to y
- C. x is a subset of y
- D. x is the image of R

Correct answer: x is related to y under R

Explanation and steps

1. The governing definition or property gives: x is related to y under R .
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is x is related to y under R .

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 12 - Theory Q12

If $(x, y) \in R$, then y is called the:

Options

- A. Image of x under R
- B. Pre-image of x

- C. Domain of R
- D. First component of x

Correct answer: Image of x under R

Explanation and steps

1. The governing definition or property gives: Image of x under R.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Image of x under R.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 13 - Theory Q13

If $(x, y) \in R$, then x is called the:

Options

- A. Pre-image of y under R
- B. Image of y
- C. Range of R
- D. Second component of y

Correct answer: Pre-image of y under R

Explanation and steps

1. The governing definition or property gives: Pre-image of y under R.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Pre-image of y under R.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 14 - Theory Q14

The domain of a relation R is the set of all:

Options

- A. First components of ordered pairs in R
- B. Second components of ordered pairs in R
- C. Elements in the co-domain
- D. Subsets of R

Correct answer: First components of ordered pairs in R

Explanation and steps

1. The governing definition or property gives: First components of ordered pairs in R.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is First components of ordered pairs in R.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 15 - Theory Q15

If R is a relation from A to B, its co-domain is:

Options

- A. B
- B. A
- C. R
- D. $A \times B$

Correct answer: B

Explanation and steps

1. The governing definition or property gives: B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 16 - Theory Q16

The range of a relation R is the set of all:

Options

- A. Second components of ordered pairs in R
- B. First components of ordered pairs in R
- C. Elements of $A \times B$ not in R
- D. Subsets of the domain

Correct answer: Second components of ordered pairs in R

Explanation and steps

1. The governing definition or property gives: Second components of ordered pairs in R.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Second components of ordered pairs in R.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 17 - Theory Q17

A relation is one-one when:

Options

- A. Distinct elements in the domain have distinct images
- B. Every element has the same image
- C. The range is a proper subset of the co-domain
- D. No element has an image

Correct answer: Distinct elements in the domain have distinct images

Explanation and steps

1. The governing definition or property gives: Distinct elements in the domain have distinct images.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Distinct elements in the domain have distinct images.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 18 - Theory Q18

A relation is many-one when:

Options

- A. Two or more domain elements have the same image
- B. Every domain element has a distinct image
- C. Every co-domain element has a pre-image
- D. The domain and range are equal

Correct answer: Two or more domain elements have the same image

Explanation and steps

1. The governing definition or property gives: Two or more domain elements have the same image.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Two or more domain elements have the same image.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 19 - Theory Q19

A relation from A to B is into when:

Options

- A. Its range is a proper subset of B
- B. Its range is equal to B
- C. Its domain is empty
- D. It is equal to $A \times B$

Correct answer: Its range is a proper subset of B

Explanation and steps

1. The governing definition or property gives: Its range is a proper subset of B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Its range is a proper subset of B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 20 - Theory Q20

A relation from A to B is onto when:

Options

- A. Every element of B is the image of some element of A
- B. Every element of A has two images
- C. Its range is a proper subset of B
- D. Its domain and co-domain are disjoint

Correct answer: Every element of B is the image of some element of A

Explanation and steps

1. The governing definition or property gives: Every element of B is the image of some element of A.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Every element of B is the image of some element of A.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 21 - Theory Q21

A binary relation on a non-empty set A is:

Options

- A. Any subset of $A \times A$

- B. Any subset of A only
- C. Only the empty set
- D. Any subset of $A \times B$ where $B \neq A$

Correct answer: Any subset of $A \times A$

Explanation and steps

1. The governing definition or property gives: Any subset of $A \times A$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Any subset of $A \times A$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 22 - Theory Q22

Which is the universal relation on A?

Options

- A. $A \times A$
- B. $\emptyset$
- C. $A \cup A$
- D. $P(A)$

Correct answer: $A \times A$

Explanation and steps

1. The governing definition or property gives: $A \times A$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A \times A$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 23 - Theory Q23

If $n(A) = m$ and $n(B) = n$, the number of relations from A to B is:

Options

- A. 2^{mn}
- B. $m \times n$
- C. $2^m + 2^n$
- D. m^n

Correct answer: 2^{mn}

Explanation and steps

1. The governing definition or property gives: 2^{mn} .
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is 2^{mn} .

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 24 - Theory Q24

A relation R on A is reflexive when:

Options

- A. $(a, a) \in R$ for every $a \in A$
- B. $(a, b) \in R$ implies $(b, a) \in R$
- C. $(a, b) \in R$ and $(b, c) \in R$ imply $(a, c) \in R$
- D. No pair (a, a) belongs to R

Correct answer: $(a, a) \in R$ for every $a \in A$

Explanation and steps

1. The governing definition or property gives: $(a, a) \in R$ for every $a \in A$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $(a, a) \in R$ for every $a \in A$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 25 - Theory Q25

A relation R on A is symmetric when:

Options

- A. $(a, b) \in R$ implies $(b, a) \in R$
- B. $(a, a) \in R$ for every $a \in A$
- C. $(a, b) \in R$ implies $a = b$
- D. $(a, b) \in R$ and $(b, c) \in R$ imply $(a, c) \in R$

Correct answer: $(a, b) \in R$ implies $(b, a) \in R$

Explanation and steps

1. The governing definition or property gives: $(a, b) \in R$ implies $(b, a) \in R$.
2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $(a, b) \in R$ implies $(b, a) \in R$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 26 - Theory Q26

A relation R on A is transitive when:

Options

- A. $(a, b) \in R$ and $(b, c) \in R$ imply $(a, c) \in R$
- B. $(a, b) \in R$ implies $(b, a) \in R$
- C. $(a, a) \in R$ for every $a \in A$
- D. Every element has exactly one image

Correct answer: $(a, b) \in R$ and $(b, c) \in R$ imply $(a, c) \in R$

Explanation and steps

1. The governing definition or property gives: $(a, b) \in R$ and $(b, c) \in R$ imply $(a, c) \in R$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $(a, b) \in R$ and $(b, c) \in R$ imply $(a, c) \in R$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 27 - Theory Q27

An equivalence relation is:

Options

- A. Reflexive, symmetric, and transitive
- B. Only reflexive and symmetric
- C. Only symmetric and transitive
- D. One-one and onto

Correct answer: Reflexive, symmetric, and transitive

Explanation and steps

1. The governing definition or property gives: Reflexive, symmetric, and transitive.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Reflexive, symmetric, and transitive.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 28 - Theory Q28

For arbitrary non-empty sets A and B, which statement is generally true?

Options

- A. $A \times B \neq B \times A$
- B. $A \times B = B \times A$ always
- C. $A \times B = A \cap B$
- D. $A \times B = A \cup B$

Correct answer: $A \times B \neq B \times A$

Explanation and steps

1. The governing definition or property gives: $A \times B \neq B \times A$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $A \times B \neq B \times A$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 29 - Theory Q29

If $n(A) = 4$ and $n(B) = 3$, enter $n(A \times B)$.

Correct answer: 12

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 12.
3. Therefore, the required answer is 12.

Question 30 - Theory Q30

If $n(A) = 2$ and $n(B) = 3$, enter the total number of relations from A to B.

Correct answer: 64

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 64.
3. Therefore, the required answer is 64.

Question 31 - Theory Q31

For $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(x, y) / y = x + 1\}$, enter $n(R)$.

Correct answer: 5

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 5.
3. Therefore, the required answer is 5.

Question 32 - Theory Q32

If $(x + 3, 2) = (4, y - 3)$, enter x .

Correct answer: 1

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 1.
3. Therefore, the required answer is 1.

Question 33 - Theory Q33

If $(x + 3, 2) = (4, y - 3)$, enter y .

Correct answer: 5

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 5.
3. Therefore, the required answer is 5.

Question 34 - Theory Q34

For $R = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$, enter $n(\text{Domain}(R))$.

Correct answer: 4

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 4.
3. Therefore, the required answer is 4.

Question 35 - Theory Q35

For $R = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$, enter $n(\text{Range}(R))$.

Correct answer: 2

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 2.
3. Therefore, the required answer is 2.